- Unwanted blocking
- Race conditions

These are huge categories, so let's divide them up a bit. First, let's look at unwanted blocking.

10.1.1. Unwanted blocking

What do I mean by unwanted blocking? First, a thread is *blocked* when it's unable to proceed because it's waiting for something. This is typically something like a mutex, a condition variable, or a future, but it could be waiting for I/O. This is a natural part of multithreaded code, but it's not always desirable—hence the problem of unwanted blocking. This leads us to the next question: why is this blocking unwanted? Typically, this is because some other thread is also waiting for the blocked thread to perform some action, and so that thread in turn is blocked. There are several variations on this theme:

- *Deadlock*—As you saw in chapter 3, in the case of deadlock one thread is waiting for another, which is in turn waiting for the first. If your threads deadlock, the tasks they're supposed to be doing won't get done. In the most visible cases, one of the threads involved is the thread responsible for the user interface, in which case the interface will cease to respond. In other cases, the interface will remain responsive, but some required task won't complete, such as a search not returning or a document not printing.
- *Livelock*—Livelock is similar to deadlock in that one thread is waiting for another, which is in turn waiting for the first. The key difference here is that the wait is not a blocking wait but an active checking loop, such as a spin lock. In serious cases, the symptoms are the same as deadlock (the app doesn't make any progress), except that the CPU usage is high because threads are still running but blocking each other. In not-so-serious cases, the livelock will eventually resolve because of the random scheduling, but there will be a long delay in the task that got livelocked, with a high CPU usage during that delay.
- *Blocking on I/O or other external input*—If your thread is blocked waiting for external input, it can't proceed, even if the waited-for input